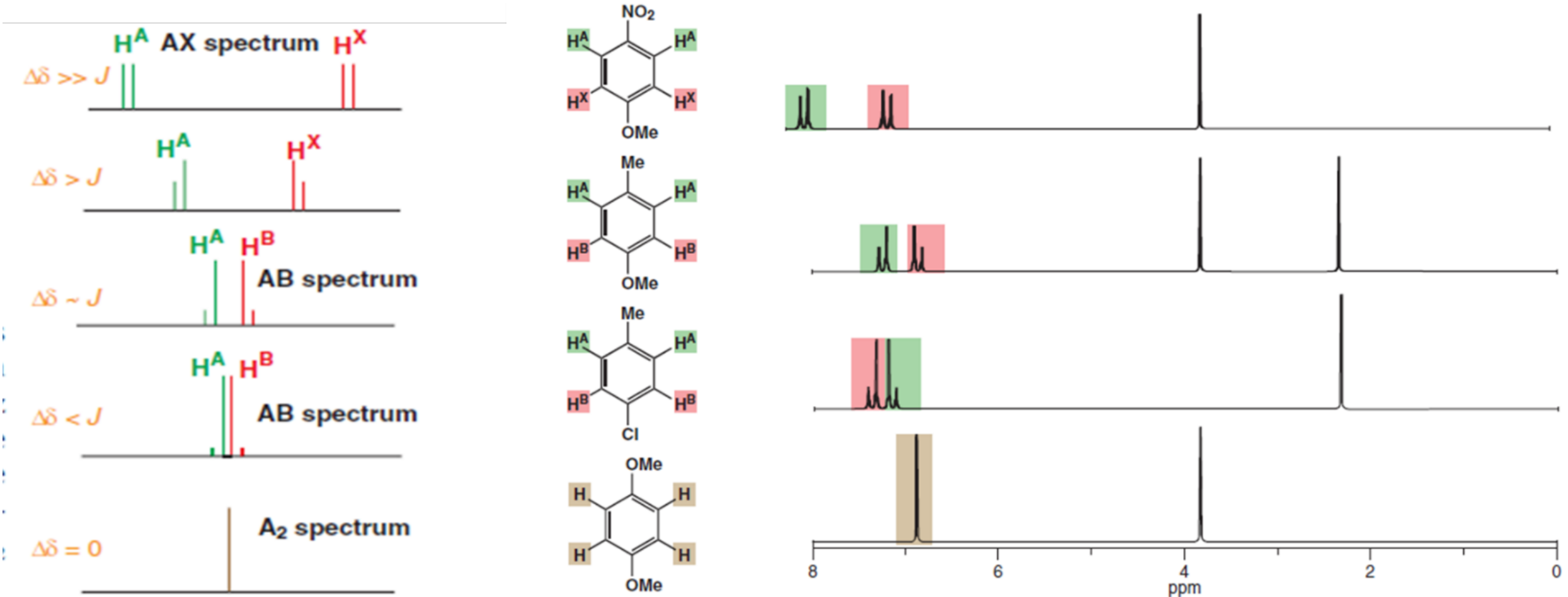
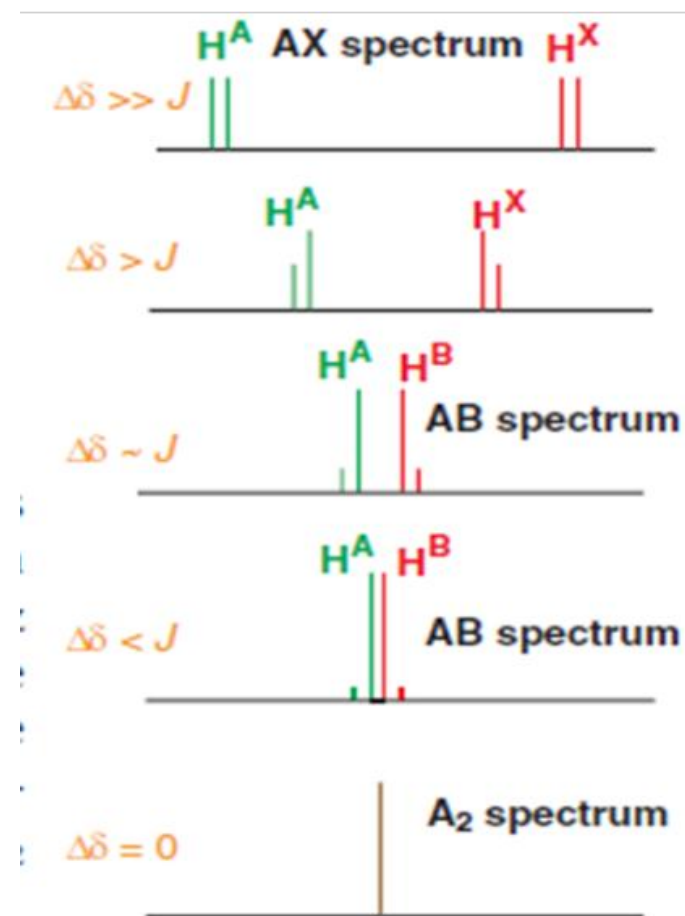
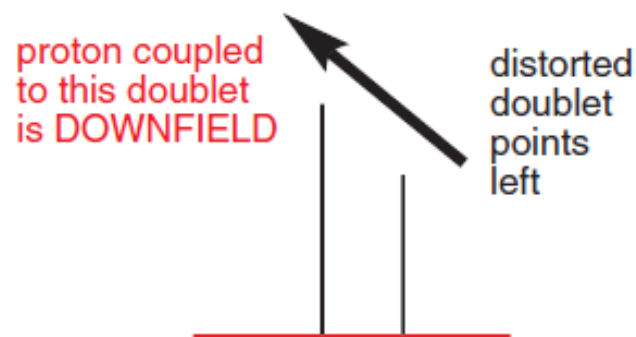
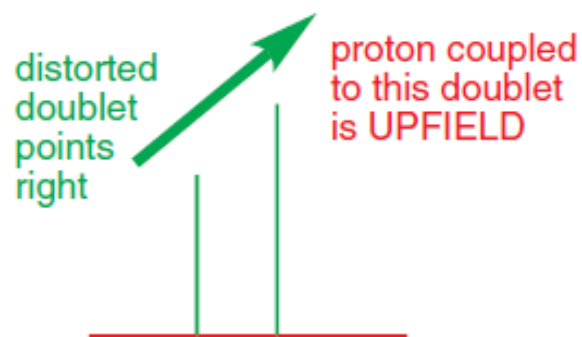
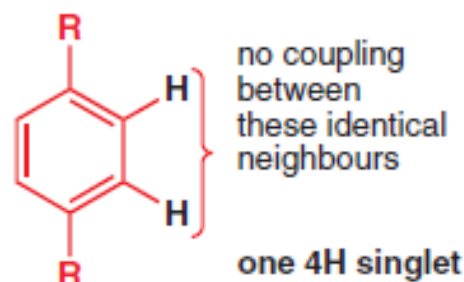
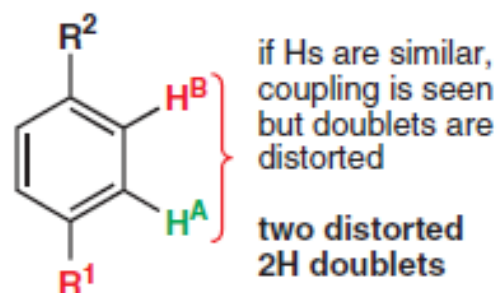
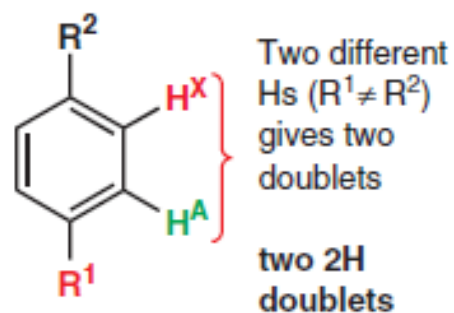


Coupling between similar protons

What happens to two similar neighbours? As two protons get closer and closer together, do the two doublets you see in the AX system suddenly collapse to the singlet of the A_2 system? You have probably guessed that they do not. The transition is gradual.



The critical factor in the shape of the peak is how the difference between the chemical shifts of the two protons ($\Delta\delta$) compares with the size of the coupling constant (J). If $\Delta\delta$ is much larger than J there is no distortion: if, say, $\Delta\delta$ is 2 ppm at 500 MHz (= 1000 Hz) and the coupling constant is a normal 7 Hz, then this condition is fulfilled and we have an AX spectrum of two 1:1 doublets. As $\Delta\delta$ approaches J in size, so the inner lines of the two doublets increase and the outer lines decrease until, when $\Delta\delta$ is zero, the outer lines vanish away altogether and we are left with the two superimposed inner lines—a singlet or an A₂ spectrum.



Spin System Notation

Each chemically different proton is given a capital letter:

AX: A and X have large chemical shift difference

AB: A and B have very small chemical shift difference

A₂: Identical chemical shifts and magnetically equivalent

AA': Chemical shift equivalent but magnetically non-equivalent

ABC: Three groups of protons with very similar chemical shifts

AMX: Three groups of protons with large differences in chemical shifts

First Order Spectra

If the splitting pattern of a multiplet is given by the $n+1$ rule, then the spectra is called as ideal or “first order” spectra. This is usually observed if the spin-coupled nuclei have very different chemical shifts (i.e. $\Delta\nu$ is large compared to J). The chemical shift value and the coupling constant J can be measured directly from the spectrum.

The first order spectra can be analysed by the following features

1. The multiplicity of a peak is given by the expression $2nI+1$, where I is the nuclear spin of the concerned nucleus and n is the total number of coupled nuclei.
2. The relative intensity of the component peaks of a multiplet is given by the coefficients of the terms in binomial expansion.
3. Centre of the multiplet gives the chemical shift of the concerned nucleus.
4. The separations of peaks in two coupled multiplets are exactly equal and correspond to the coupling constant.

It is well known that the chemical shift measured in Hz depends on the applied magnetic field and the value of $\Delta\nu$ increases as the field strength increases. The value of coupling constant J remains constant and hence the value of $\Delta\nu/J$ will increase on increasing the field strength. Therefore, at high field strength, most ^1H NMR spectra become first order. It is important to notice that high field NMR instruments give better resolution and relatively easily interpretable spectra.

Second Order Spectra

If the coupled nuclei have approximately same chemical shifts and their chemical shift difference is comparable to the coupling constant between them, the first order splitting rule ($2nI+1$) is not applicable. The splitting pattern in a multiplet is distorted and the bands are no longer symmetrical. In fact, signal splitting disappears if the chemical shifts of the coupled nuclei are the same. Such types of spectra are known as second order spectra. It is difficult to measure the chemical shift and coupling constants values simply by just analysing the spectra.

As the value of $\Delta\nu/J$ decreases, the coupled multiplets approach each other and the inner lines increase in intensity and the outer peaks decrease. In some extreme cases the outer lines may become too weak to be observed and hence the resulting multiplet is wrongly interpreted. Since there is leaning of coupled multiplets towards one another and hence the chemical shift values cannot be given by the centre of the multiplet. In such cases the chemical shift is given by the centre of gravity.

$\Delta\nu/J \geq 10$: first order; $\Delta\nu/J \leq 10$: second order

100. Reaction of styrene ($\text{PhCH}=\text{CH}_2$) with HBr gives a mixture of regioisomers **A** (major) and **B** (minor). The ^1H NMR spectrum of the mixture shows four signals, amongst others, at δ 5.17, 3.53, 3.15 and 2.00 ppm with relative integration of 2:1:1:6, respectively. The molar ratio of **A** and **B** is

1. An organic compound ($\text{C}_7\text{H}_{12}\text{O}_2$) exhibited the following data in the ^1H NMR spectrum.

δ 7.10 (1H, dt, $J = 16$ Hz and 7.2 Hz), 5.90 (1H, dt, $J = 16$ Hz and 2 Hz), 4.1 (2H, q, $J = 7.2$ Hz), 2.10 (2H, m), 1.25 (3H, t, $J = 7.2$ Hz), 0.90 (3H, t, $J = 7.2$ Hz) ppm. (CSIR NET)

Q2. Determine the correct structure of a compound having molecular formula $\text{C}_6\text{H}_{11}\text{BrO}_2$, based on the following characteristic spectral data:

IR: 1736 cm^{-1}

^1H NMR (δ , ppm): 3.59 (s, 3H), 3.32 (t, 2H), 2.25 (t, 2H), 1.85–1.75 (m, 2H), 1.73–1.62 (m, 2H)

^{13}C NMR (δ , ppm): 174.0, 51.0, 32.9, 32.8, 31.0, 23.0

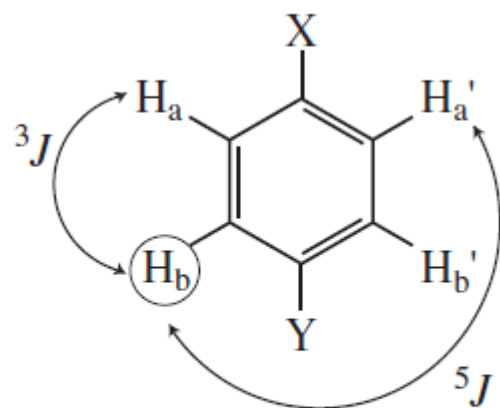
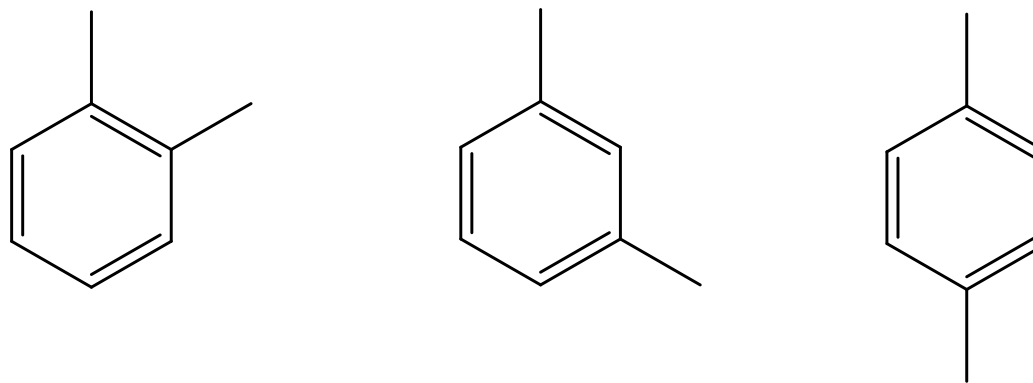
(CSIR NET)

Q. Isomeric compounds **A** and **B** have the composition $C_{11}H_{12}O_4$. Deduce structures for **A** and **B** from the spectral data (in the proton NMR, ortho and meta couplings are visible). **Compound A**: IR (KBr): 1720 (s), 1240 (s) cm^{-1} , 1H NMR ($CDCl_3$): δ 2.46 (s, 3H), 3.94 (s, 6H), 8.05 (d, J 2 Hz, 2H), 8.49 (t, J 2 Hz, 1H); **Compound B**: IR (Nujol[®]): 1720 (s), 1245 (s) cm^{-1} , 1H NMR ($CDCl_3$): δ 2.63 (s, 3H), 3.91 (s, 6H), 7.28 (d, J 8 Hz, 1H), 8.00 (dd, J 8 Hz, 2 Hz; 1H), 8.52 (d, J 2 Hz, 1H). Assign all protons to the appropriate peaks.

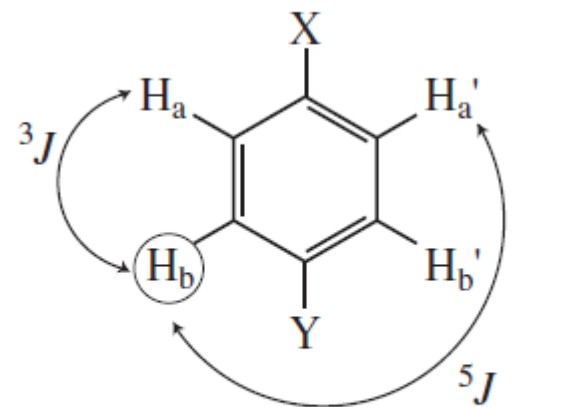
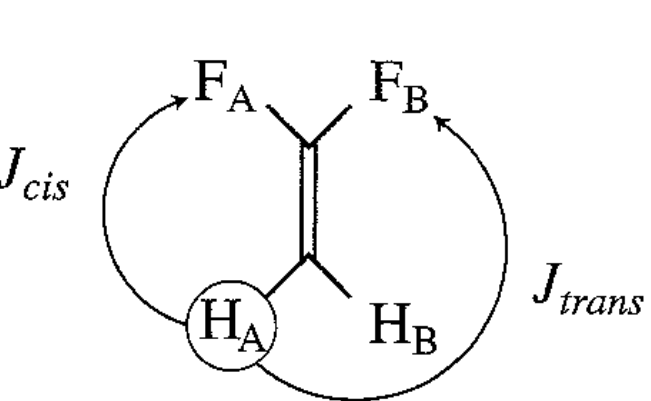
$$DBE: C - (H/2) - (X/2) + (N/2) + 1$$

Magnetic Equivalence

What about xylenes?



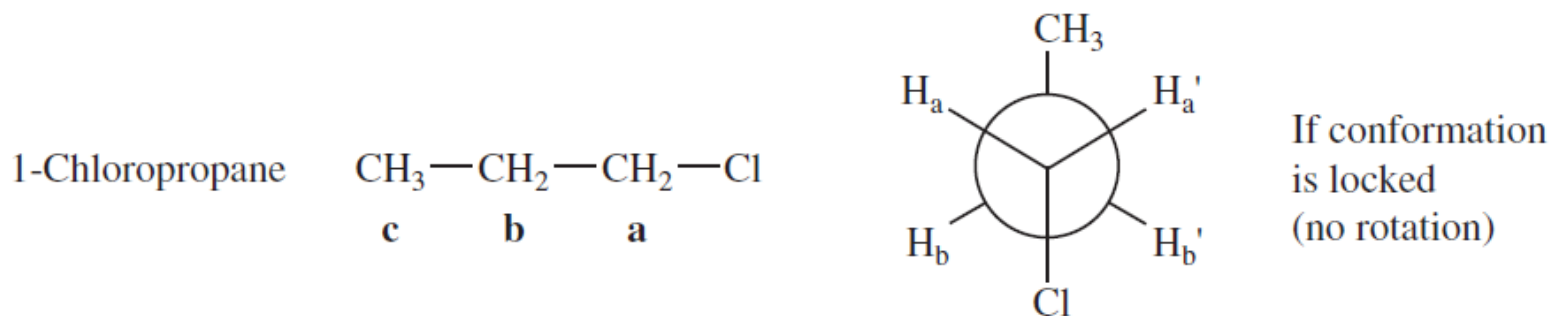
para-Disubstituted benzene



para-Disubstituted benzene

Are protons H_A and H_B magnetically equivalent?

- No, they are chemical shift equivalent but magnetically non-equivalent
- However, only a singlet is observed in the NMR spectra the dichloro, dibromo and diiodo derivatives



Now consider 1-chloropropane. The hydrogens within a group (those on C1, C2, and C3) are isochronous, but each group is on a different carbon, and as a result, each *group* of hydrogens has a different chemical shift. The hydrogens in each group experience identical *average* magnetic environments, mainly because of free rotation, and are magnetically equivalent. Furthermore, also because of rotation, the hydrogens in each group are equally coupled to the hydrogens in the other groups. If we consider the two hydrogens on C2, H_b and H_b' and pick any other hydrogen on either C1 or C3, both H_b and H_b' will have the same coupling constant to the designated hydrogen. Without free rotation (see the preceding illustration) there would be no magnetic equivalence. Because of the fixed unequal dihedral angles ($H_a|Cl|Cl|C_b$ versus $H_a|Cl|Cl|H_b'$), J_{ab} and $J_{ab'}$ would not be the same. Free rotation can be slowed or stopped by lowering the temperature, in which case H_b and H_b' would become magnetically inequivalent. This type of magnetic inequivalence is often seen in 1,2-disubstituted ethane groups in which the substituents have sufficient steric bulk to hinder free rotation around the C|C axis enough that it becomes slow on the NMR time-scale.

Why we don't see coupling of protons with halogens other than F?

Chlorine ($I = 3/2$)

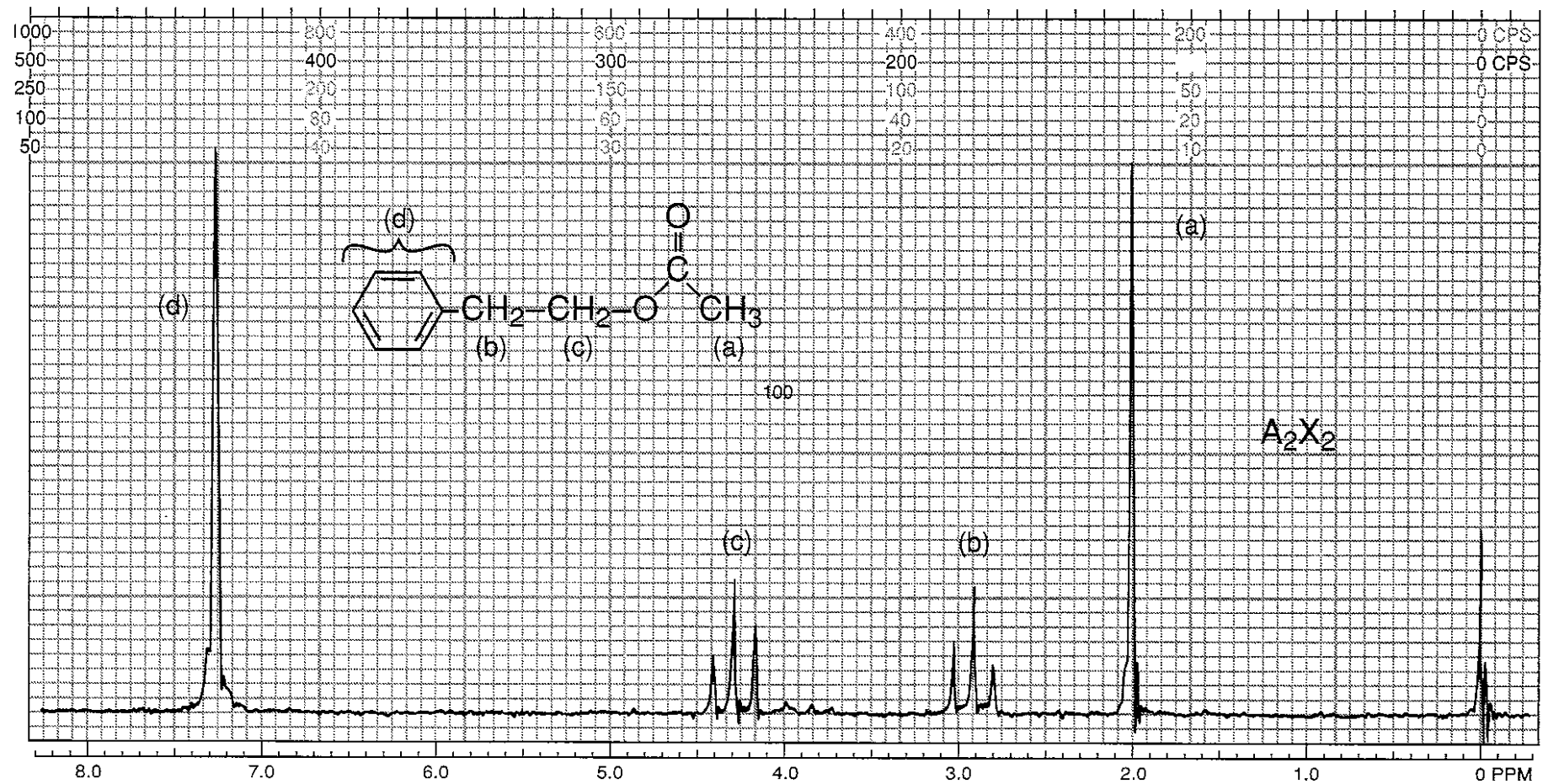
Bromine ($I = 3/2$)

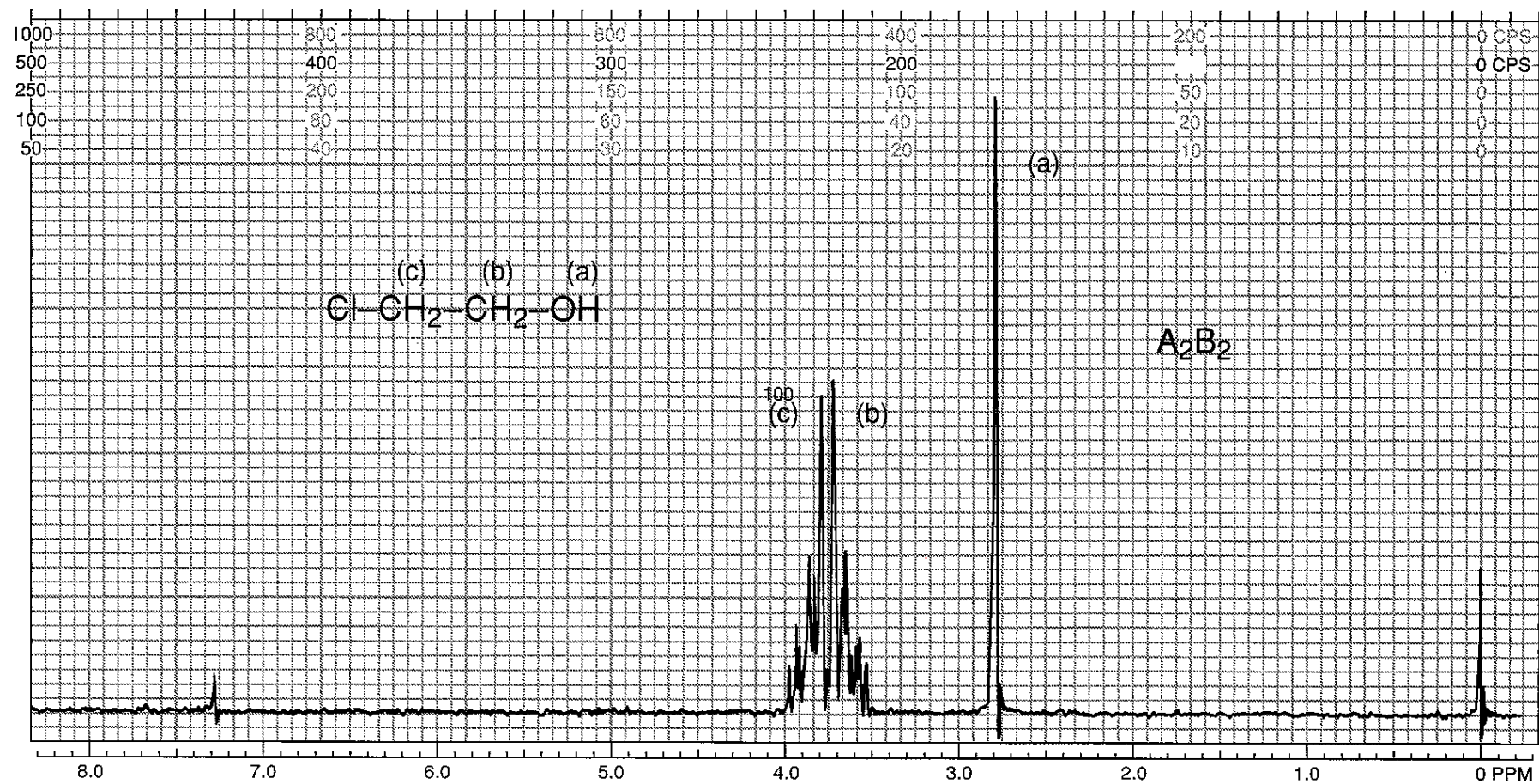
Iodine ($I = 5/2$)

➤ These halogens are too large with diffused electron cloud to transmit any spin information

➤ $^2J_{\text{HX}}, ^3J_{\text{HX}} = 0$ for Cl, Br and I

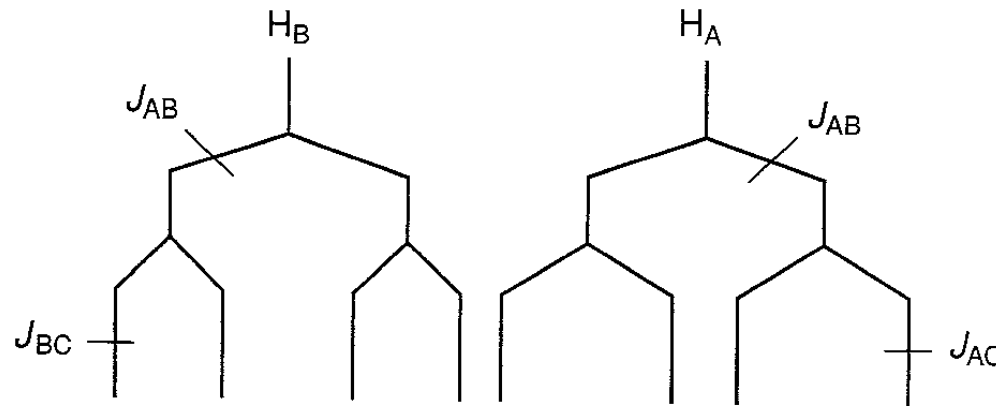
➤ Only fluorine among these halogens undergoes coupling





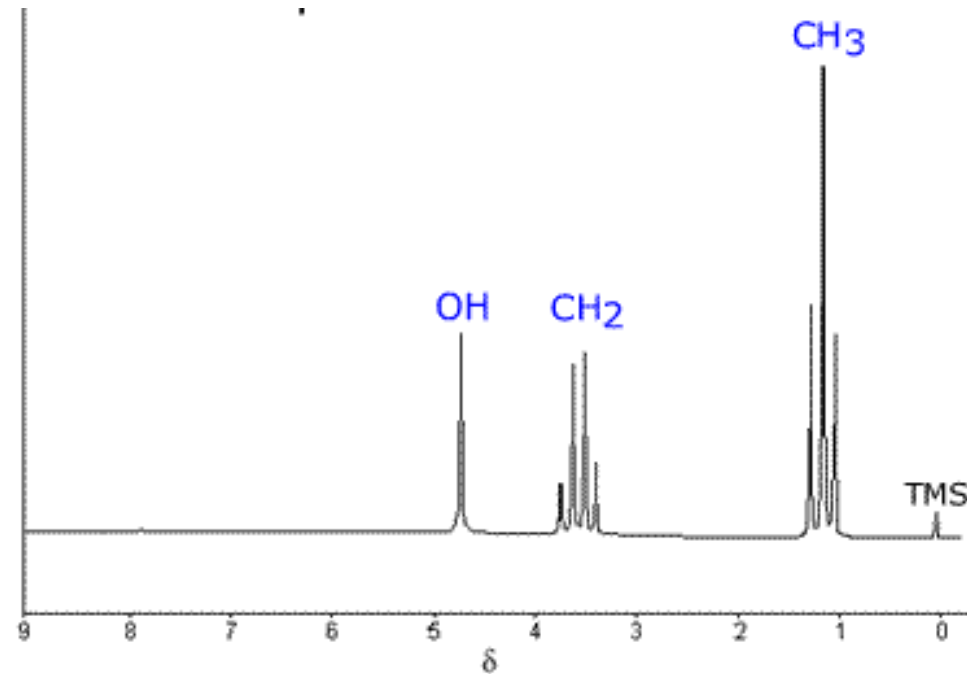
First Order Spectra

- (n+1) rule
- Splitting Tree diagram



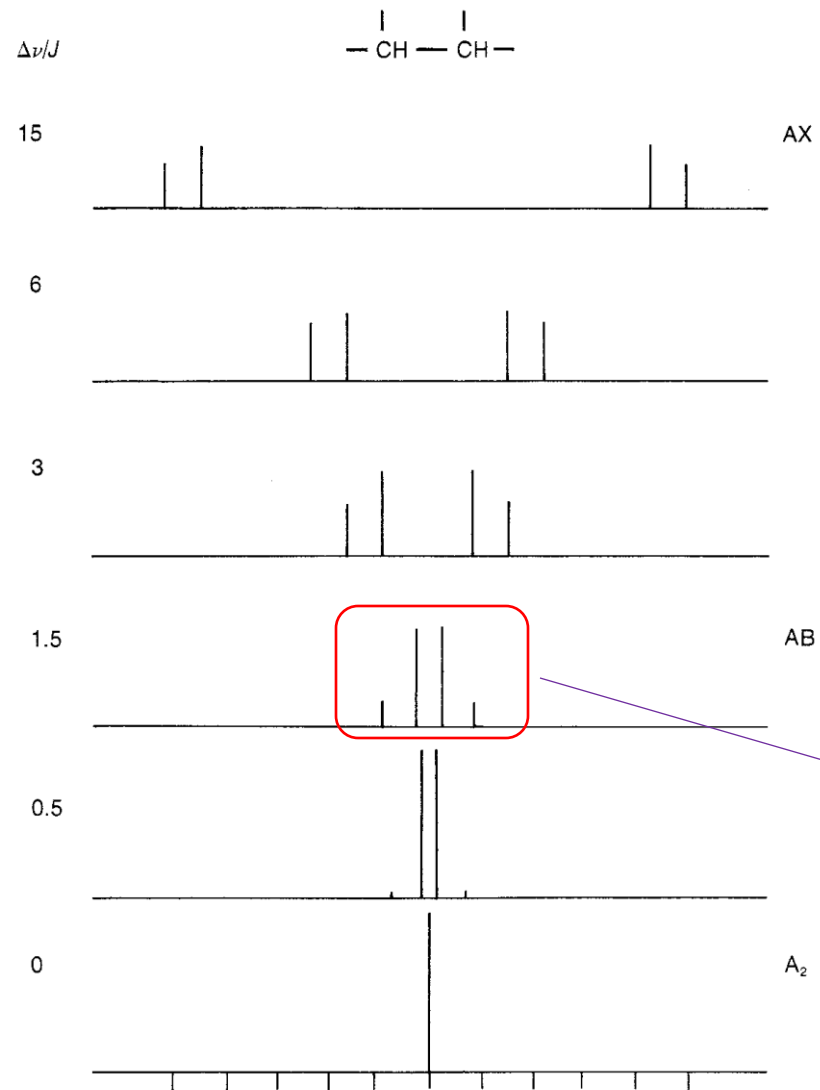
- Spectra that can be interpreted using (n+1) rule or simple graphical analysis (splitting trees) are known as *first order spectra*

NMR spectra of EtOH



First Order Spectra

➤ *Second order spectra* are generally observed when the chemical shift difference between two groups of protons is comparable to the coupling constant

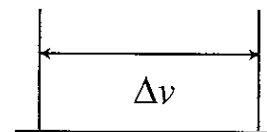
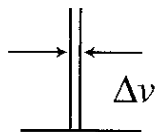
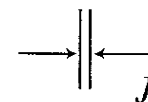
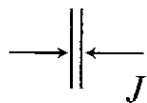


AX: Large difference in chemical shifts

AB: Small difference in chemical shifts

AB quartet

A_2 : No difference in chemical shifts

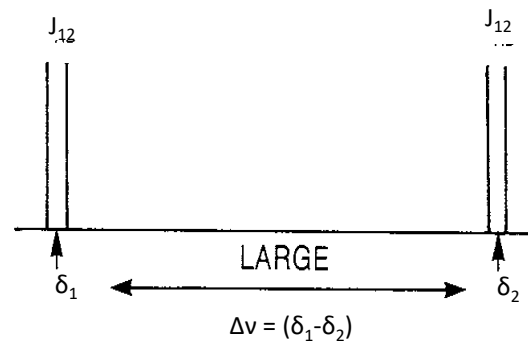
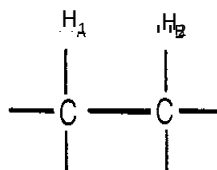


Strong coupling, second-order spectra
($\Delta\nu/J$ small)

Weak coupling, first-order spectra
($\Delta\nu/J$ large)

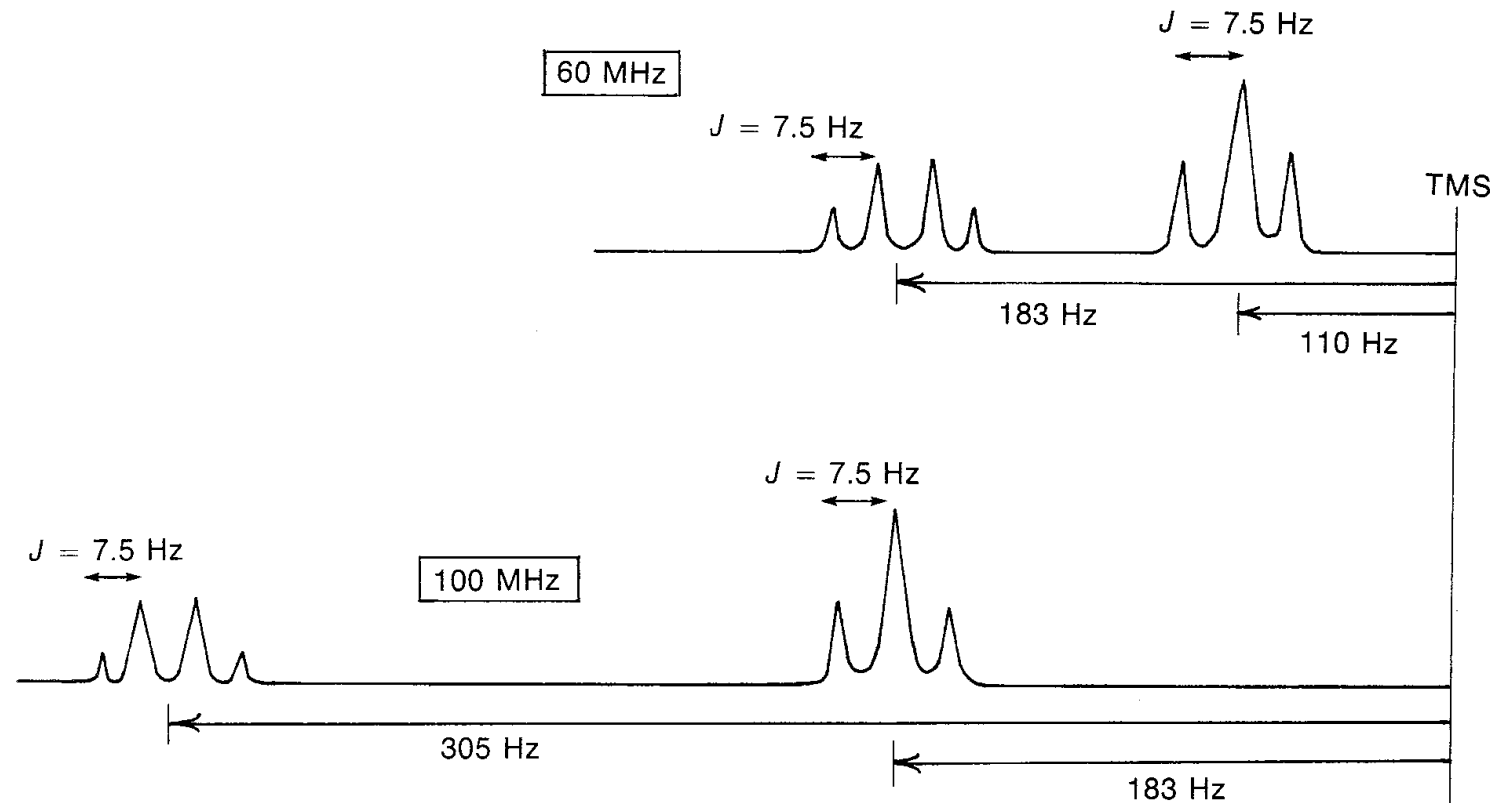
□ $\Delta\nu/J > 10$, weakly coupled system: **first order spectra**

□ $\Delta\nu/J \sim 1$, strongly coupled system: **second order spectra**

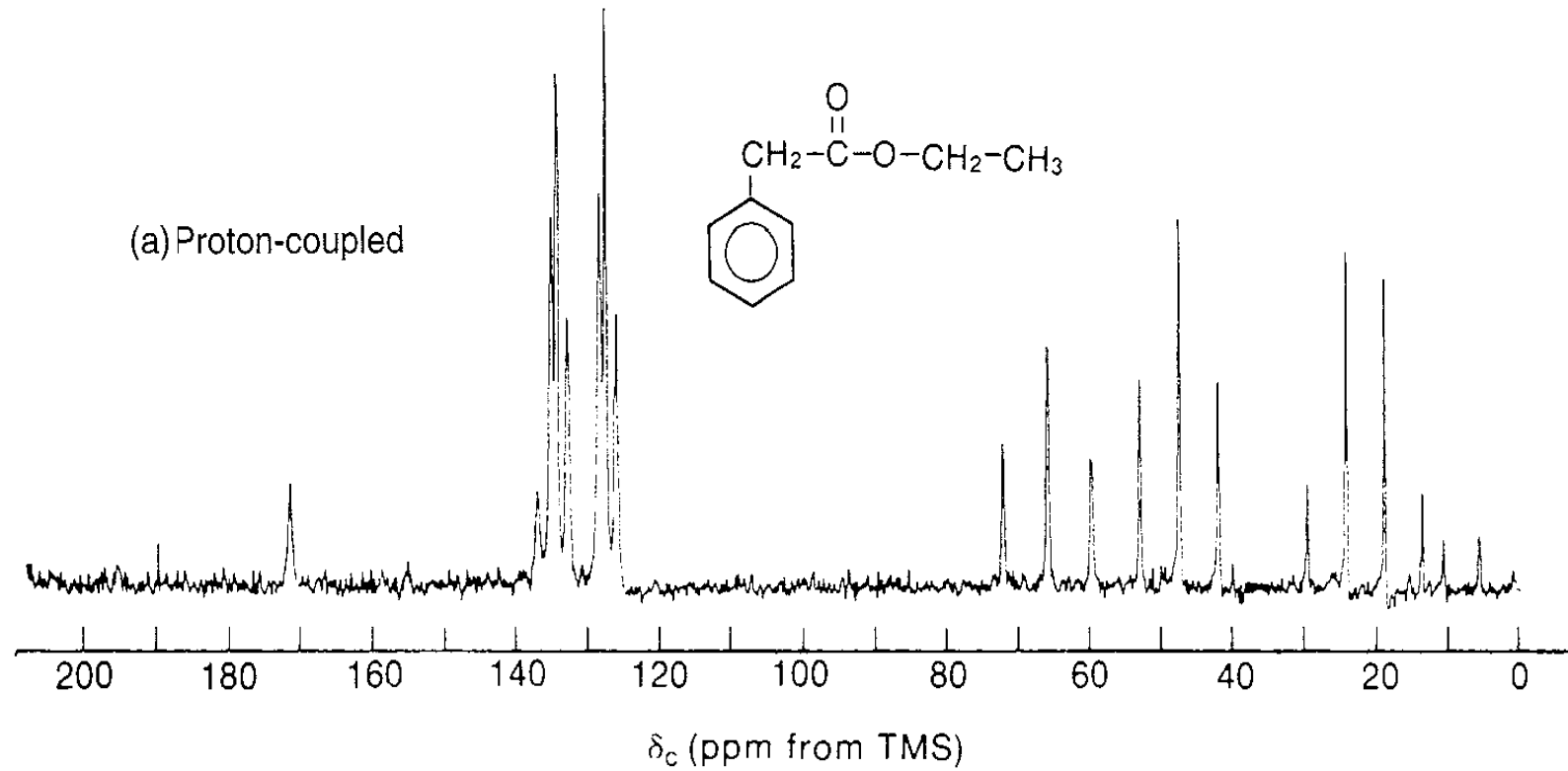


➤ Resonance frequency of protons is dependent on the magnetic field

➤ *The coupling J is independent of field strength*



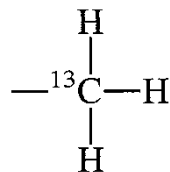
Proton Coupled Spectra



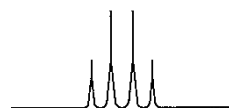
- ✓ The signal intensity is typically low in ^{13}C NMR
- ✓ Coupling to protons make the intensity of a peak even lower

Proton Coupled ^{13}C spectra: Splitting of ^{13}C Signals

3 protons

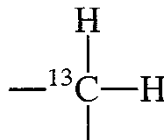


$$\begin{aligned} n + 1 &= 3 + 1 \\ &= 4 \end{aligned}$$

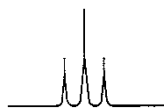


Methyl
carbon

2 protons

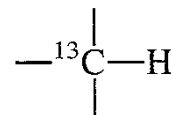


$$n + 1 = 3$$

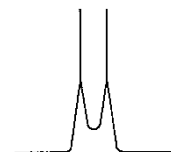


Methylene
carbon

1 proton

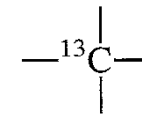


$$n + 1 = 2$$

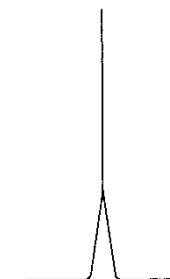


Methine
carbon

0 protons



$$n + 1 = 1$$



Quaternary
carbon

✓ The coupling constants (1J) are typically 100-200 Hz